

Two Standard DSGC Model Beds: A Side-by-Side HH-Equation Reference (v2)

Task t0071_t0070_synaptic_eqs_pdf

2026-05-01

Note

This document is the **v2 writeup** of the project's two standard DSGC model beds and **supersedes** t0070's `results/results_detailed.md`. Two gaps in the v1 writeup are addressed here:

1. **Synaptic-current equations were missing.** The Hodgkin-Huxley membrane equation and every active conductance were given as explicit $I = g \cdot \dots \cdot (V - E)$ blocks, but the synaptic currents (AMPA, NMDA, GABA, ACh in Bed A; Exp2Syn ACh, Exp2Syn GABA in Bed B) were described in prose and parameter tables only. This v2 adds the canonical

$$I_{\text{syn}}(t, V) = g_{\text{syn}}(t, V) \cdot (V - E_{\text{syn}})$$

block under each bed's "Synaptic excitation" and "Synaptic inhibition" subsections, with the state-variable evolution and discrete release rule clearly stated. Five new equation blocks total: Bed A BIPsyn AMPA, Bed A BIPsyn NMDA, Bed A SACinhibsyn GABA, Bed A SACexcsyn ACh, and Bed B Exp2Syn ACh + Exp2Syn GABA.

2. **No typeset PDF.** v1 was markdown only. This v2 ships `results/results_detailed.typ` (Typst source) and `results/results_detailed.pdf` (typeset PDF) so equations render in proper math fonts. Every equation in the v1 fenced text blocks has been rewritten in math syntax so downstream renderers typeset them correctly.

No biophysical parameter value has changed between t0070 and this v2. Every numerical value carries the same `code/<file>:L<line>` citation; only the surrounding markup has changed and the five new equation blocks have been added.

Summary

This task supersedes t0070's research-paper writeup of the project's two canonical direction-selective ganglion cell (DSGC) model substrates, adding the missing synaptic-current equation blocks and a typeset PDF. **Bed A** is the deposited Poleg-Polsky 2016 ON-OFF DRD4 DSGC ported in t0008 (ModelDB 189347, library `modeldb_189347_dsgc`); it is the substrate for t0020, t0065, t0067, t0068, and t0069. **Bed B** is the de Rosenroll 2026 DSGC ported in t0024 (library `de_rosenroll_2026_dsgc`); it is the substrate for t0066. Each bed is presented in parallel structure: Morphology → canonical Hodgkin-Huxley membrane equation → conductance table → synaptic excitation (PD vs ND) → synaptic inhibition (PD vs ND) → differences from the original paper. A side-by-side comparison table at the end summarises the headline differences across 18 dimensions. Every numerical parameter carries a `code/<file>:line` citation back to the committed source so the document is fully auditable.

Methodology

- **Machine:** local Windows 11 workstation; no remote compute, no GPU, no NEURON simulations.
- **Runtime:** total 2 hours of equation transcription, Typst authoring, and PDF rendering. Typst compile under 5 seconds.
- **Methods:** documentation-only correction task. No new experiments were run. Every parameter quoted is the same value cited in t0070, traced back to a specific line of the committed library source.
- **Schematic figures:** four PNGs in `results/images/` are reused verbatim from t0070; copied with `shutil.copyfile` and not regenerated.
- **Equation typesetting:** every equation is in math syntax. Inline math uses `$...$` (no whitespace inside delimiters) and display math uses `$ ... $` (with whitespace) per Typst convention.

Bed A: Poleg-Polsky 2016 (t0008 + t0020)

Bed A wraps the deposited Poleg-Polsky & Diamond 2016 ON-OFF DRD4 DSGC from ModelDB 189347 with no modification to the morphology or kinetics. The library asset is `tasks/t0008_port_modeldb_189347/assets/library/modeldb_189347_dsgc/`.

Morphology

Bed A: Poleg-Polsky 2016 (t0008) — 1 soma + 350 dends, 282 ON / 68 OFF, 846 point processes

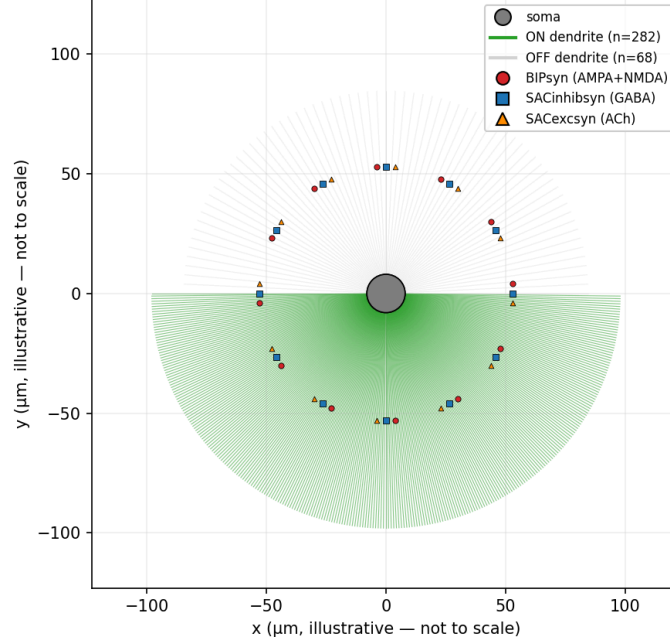


Figure 1: Figure A1 (illustrative; not to scale): 1 soma + 350 dendrite sections; 282 ON-typed dendrites bear the synaptic triple (BIPsyn AMPA + NMDA bipolar drive, SACinhibsyn GABA, SACexcsyn ACh); the OFF dendrites bear no synapses.

The cell is built by the deposited HOC template `RGModel.hoc` (11 861 lines). Key facts:

- **Section count:** 1 soma + 350 `dend[i]` sections (`RGModel.hoc:L14`).
- **Compartment discretisation:** every section has $n_{\text{seg}} = 1$ (`RGModel.hoc:L11818`).
- **Per-dendrite diameter:** $\text{diam} = 0.5 + 2.58 \cdot \exp\left(-\frac{d(0.5)-10}{10}\right) \mu\text{m}$ (`RGModel.hoc:L11814`).
- **ON / OFF sort:** 282 ON dendrites, 68 OFF (`RGModel.hoc:L11801-L11803`).
- **Synapse placement:** 846 point processes per cell (`RGModel.hoc:L11825-L11851`).
- **Passive cable:** $R_a = 100 \Omega \cdot \text{cm}$, $c_m = 1 \mu\text{F}/\text{cm}^2$, $\text{celsius} = 32^\circ\text{C}$, $V_{\text{init}} = -65 \text{ mV}$.

Membrane equation

The Hodgkin-Huxley membrane equation governs each compartment of Bed A:

$$C_m \frac{dV}{dt} = - \sum_i I_i - I_{\text{syn}} - I_{\text{inj}}$$

The active currents $\sum_i I_i$ enumerated for Bed A (taken from `HHst.mod:L190-L222` BREAKPOINT block) are:

$$I_{\text{Na}} = g_{\text{Na}} \cdot m^3 \cdot h \cdot (V - E_{\text{Na}}) \quad (\text{HHst Na fast})$$

$$I_{\text{Kdr}} = g_{\text{Kdr}} \cdot n^4 \cdot (V - E_{\text{K}}) \quad (\text{HHst K delayed rectifier})$$

$$I_{\text{Km}} = g_{\text{Km}} \cdot n_m \cdot (V - E_{\text{K}}) \quad (\text{HHst K M-type})$$

$$I_{\text{leak}} = g_{\text{leak}} \cdot (V - E_{\text{leak}}) \quad (\text{HHst leak; zleak noise term off})$$

The `HHst.mod` source also defines $I_{\text{CaL}} = g_{\text{CaL}} \cdot l_m^2 \cdot l_h \cdot (V - E_{\text{Ca}})$ and $I_{\text{CaT}} = g_{\text{CaT}} \cdot t_m^2 \cdot t_h \cdot (V - E_{\text{Ca}})$ (`HHst.mod:L208-L209`), but Bed A **explicitly zeros** these on every section in `init_active()` (`main.hoc:L155-L156`). A global voltage shift $\Delta V_{\text{shift_HHst}} = -4 \text{ mV}$ is applied to every gating variable.

Conductance table (Bed A)

Channel	Gating	gbar (S/cm ²)	E _{rev} (mV)	Notes	Source
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Na (soma)	m^3h	0.4	+60	HHst Na fast	main.hoc:L148
Na (dend)	m^3h	2e-4	+60	HHst Na fast	main.hoc:L152
Kdr (soma)	n^4	0.07	-90	HHst K delayed rectifier	main.hoc:L149
Kdr (dend)	n^4	7e-3	-90		main.hoc:L153
Km (soma)	n_m	5e-4	-90	HHst K M-type, tau _{km} = 1	main.hoc:L150
Km (dend)	n_m	0	-90	zeroed	main.hoc:L154
Leak (passive)	-	5e-5	-60	pas mode	main.hoc:L161
Leak (active)	-	5.5e-4	-60	gleak_HHst	main.hoc:L161
CaL	$l_m^2 l_h$	0	E_{Ca}	zeroed by init_active	main.hoc:L155, L302
CaT	$t_m^2 t_h$	0	E_{Ca}	zeroed by init_active	main.hoc:L156, L303

Synaptic excitation (PD vs ND)

Bed A synaptic conductance schematic — gabaMOD modulation-envelope swap (t0020 / t0065 PD/ND convention)

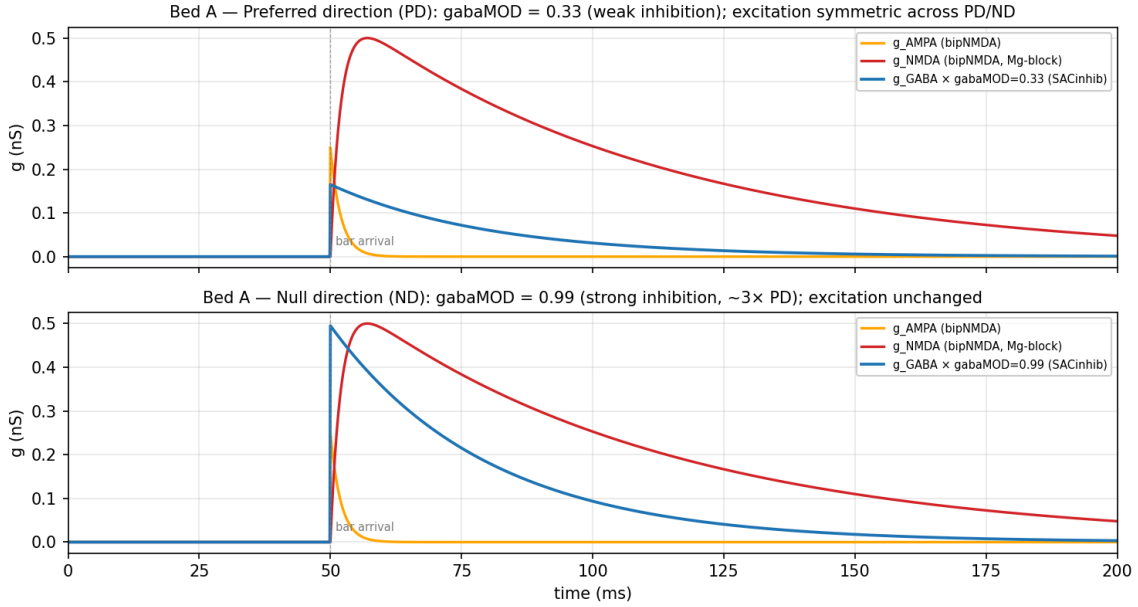


Figure 2: Figure A2 (illustrative; analytically computed, not a NEURON trace): top row plots g_{AMPA} , g_{NMDA} , and $g_{GABA} \cdot gabaMOD = 0.33$ (PD); bottom row uses $gabaMOD = 0.99$ (ND). Bipolar excitation is unchanged across PD and ND.

Bed A's excitatory drive is implemented as a single `bipNMDA` POINT_PROCESS combining AMPA and voltage-dependent NMDA (`bipolarNMDA.mod`). One `BIPsyn` instance is placed at the midpoint of every ON dendrite (282 instances total).

AMPA component — canonical synaptic-current block

The AMPA conductance is a single-decay state and the current obeys the canonical ohmic form (`bipolarNMDA.mod:L103`):

$$I_{AMPA}(t, V) = 10^{-3} g_{AMPA}(t) (V - E_{AMPA}), \quad E_{AMPA} = 0 \text{ mV}$$

with state-variable evolution and per-release update (`bipolarNMDA.mod:L156, L143`):

$$\frac{dg_{AMPA}}{dt} = -\frac{g_{AMPA}}{\tau_{AMPA}}, \quad \tau_{AMPA} = 2.0 \text{ ms}$$

$$\text{on each release event } k: \quad g_{AMPA} \leftarrow g_{AMPA} + r_k \bar{g}_{AMPA}^{\text{single}}, \quad \bar{g}_{AMPA}^{\text{single}} = 0.25 \text{ nS}$$

The factor 10^{-3} converts g_{AMPA} from nS to μS so that I is in nA when V is in mV (NEURON convention). $r_k \in \{0, 1\}$ is the per-vesicle Bernoulli release outcome.

NMDA component – bi-exponential rise/decay $\times$ Mg block

The NMDA conductance is the difference of two exponentials gated by a Jahr-Stevens-style Mg block (`bipolarNMDA.mod:L102, L104`):

$$g_{\text{NMDA}}(t, V) = \frac{A(t) - B(t)}{1 + n \cdot \exp(-\gamma V_{\text{loc}})}$$

$$I_{\text{NMDA}}(t, V) = 10^{-3} g_{\text{NMDA}}(t, V) (V - E_{\text{NMDA}}), \quad E_{\text{NMDA}} = 0 \text{ mV}$$

with $V_{\text{loc}} = V(1 - V_{\text{off}}) + V_{\text{set}} V_{\text{off}}$ ($V_{\text{off}} = 0$ uses the postsynaptic voltage; $V_{\text{off}} = 1$ clamps to $V_{\text{set}} = -43 \text{ mV}$). Bi-exponential states (`bipolarNMDA.mod DERIVATIVE` block):

$$\frac{dA}{dt} = -\frac{A}{\tau_{1,\text{NMDA}}}, \quad \frac{dB}{dt} = -\frac{B}{\tau_{2,\text{NMDA}}}$$

$$\text{on each release event } k: \quad A \leftarrow A + r_k \bar{g}_{\text{NMDA}}^{\text{single}}, \quad B \leftarrow B + r_k \bar{g}_{\text{NMDA}}^{\text{single}}$$

with $\tau_{1,\text{NMDA}} = 60 \text{ ms}$ (decay), $\tau_{2,\text{NMDA}} = 2 \text{ ms}$ (rise), $\bar{g}_{\text{NMDA}}^{\text{single}} = 0.5 \text{ nS}$, $n = 0.30 \text{ mM}^{-1}$, $\gamma = 0.07 \text{ mV}^{-1}$.

Vesicular release model (shared by AMPA and NMDA)

Vesicle release is updated every $\Delta t_{\text{rel}} = 1 \text{ ms}$ (`bipolarNMDA.mod:L81`). The presynaptic envelope V_{pre} tracks the bar-driven input V_{inf} via a low-pass filter (`bipolarNMDA.mod:L157`):

$$\frac{dV_{\text{pre}}}{dt} = \frac{V_{\text{inf}} - V_{\text{pre}}}{\tau_V}, \quad \tau_V = 30 \text{ /ms}$$

Each release-tick (`bipolarNMDA.mod:L131-L152`), the per-vesicle release probability is set to $s_{\infty} = \frac{V_{\text{pre}}}{100}$, and over the available pool of `numves` vesicles each is released independently with probability s_{∞} . The vesicle pool has `maxves` = 10 and replenishes at rate `newves` = 0.002 ms^{-1} (project override of the `bipolarNMDA.mod:L23` default of 0.01 ms^{-1}).

SACexcsyn (cholinergic) – canonical synaptic-current block

A parallel cholinergic SAC pathway is also active: `SACexcsyn` (one per ON dendrite). Its kinetic form is identical to the `SACinhibsyn` GABA mechanism:

$$I_{\text{ACh}}(t, V) = 10^{-3} g_{\text{ACh}}(t) (V - E_{\text{ACh}}), \quad E_{\text{ACh}} = 0 \text{ mV}$$

$$\frac{dg_{\text{ACh}}}{dt} = -\frac{g_{\text{ACh}}}{\tau_{\text{ACh}}}, \quad \tau_{\text{ACh}} = 3 \text{ ms}$$

$$\text{on release event } k: \quad g_{\text{ACh}} \leftarrow g_{\text{ACh}} + r_k \bar{g}_{\text{ACh}}^{\text{single}}, \quad \bar{g}_{\text{ACh}}^{\text{single}} = 0.5 \text{ nS}$$

Driven by the same Bernoulli release model as `bipNMDA`, with $s_{\infty} = \frac{V_{\text{pre}}}{100}$ and the modulating scalar `achMOD` = 0.25 multiplied into the presynaptic envelope (`main.hoc:L47`).

Synaptic inhibition (PD vs ND)

Bed A's inhibition is a single `SACinhib` POINT_PROCESS (`SAC2RGCinhib.mod`, 95 lines). One `SACinhibsyn` instance per ON dendrite (282 instances).

SACinhibsyn (GABAergic) – canonical synaptic-current block

The conductance is a single-decay state with vesicular release (`SAC2RGCinhib.mod:L55, L92, L83`):

$$I_{\text{GABA}}(t, V) = 10^{-3} g_{\text{GABA}}(t) (V - E_{\text{GABA}}), \quad E_{\text{GABA}} = -60 \text{ mV}$$

$$\frac{dg_{\text{GABA}}}{dt} = -\frac{g_{\text{GABA}}}{\tau_{\text{GABA}}}, \quad \tau_{\text{GABA}} = 30 \text{ ms}$$

on release event k : $g_{\text{GABA}} \leftarrow g_{\text{GABA}} + r_k \bar{g}_{\text{GABA}}^{\text{single}}$, $\bar{g}_{\text{GABA}}^{\text{single}} = 0.5 \text{ nS}$

The release model uses the same Bernoulli-over-vesicles structure. Direction selectivity enters via the multiplicative `gabaMOD` scalar applied to V_{inf} in `placeBIP()` (`dsgc_model.hoc:L242-L243`):

`mulnoise.fill( $V_{\text{ampT}} \cdot \text{gabaMOD}, \dots$ )  $\Rightarrow$  noisevecSACI[ $i$ ].mul(mulnoise)  $\Rightarrow$  SACinhibsyn[ $i$ ]. $V_{\text{inf}}$`

so a `gabaMOD` = 0.99 trial scales V_{inf} (and thus s_{∞} , and thus the per-vesicle release probability) by $3\times$ relative to a `gabaMOD` = 0.33 trial. The per-vesicle peak conductance $\bar{g}_{\text{GABA}}^{\text{single}} = 0.5 \text{ nS}$ is unchanged across PD and ND.

Trial	gabaMOD	Effective release-probability scaling	Source
PD (preferred)	0.33	weak inhibition; s_{∞} envelope 33% of peak	<code>constants.py:L37</code>
ND (null)	0.99	strong inhibition; s_{∞} envelope 99% of peak ($3\times$ PD)	<code>constants.py:L38</code>

Bed B: de Rosenroll 2026 (t0024)

Bed B wraps a subset of `geoffder/ds-circuit-ei-microarchitecture` at commit `a23f642aa6557a23a51bf76f51e420e8149773fa`, vendored as `tasks/t0024_port_de_rolenroll_2026_dsgc/assets/library/de_rolenroll_2026_dsgc/`. The cell template is inherited from the same Poleg-Polsky bundled cell as Bed A, but the active-channel densities are tier-stratified, the HHst variant is the noiseless deterministic derivative, and synapses are reimplemented in Python on top of `Exp2Syn`.

Morphology

Bed B: de Rosenroll 2026 (t0024) — 1 soma + 350 dends, ~ 10 primary / ~ 163 mid / ~ 177 terminal

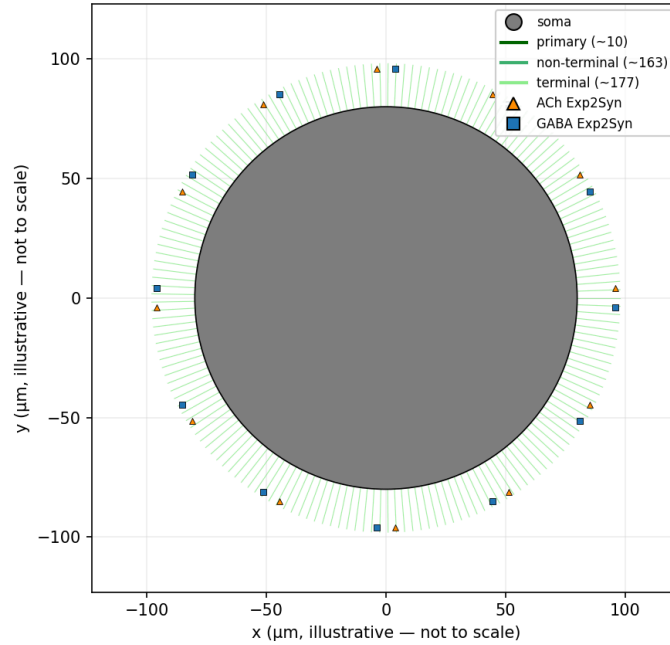


Figure 3: Figure B1: 1 soma + 350 dendrites discretised into three Python-derived tiers: 10 primary dendrites (darker green), 163 non-terminal mid dendrites (medium green), and 177 terminal dendrites (light green) bearing one Exp2Syn ACh + one Exp2Syn GABA each.

The cell uses the same 350-section template as Bed A, reimplemented as `RGCmodelGD.hoc` (11 830 lines). Key facts:

- Section count:** 1 soma + 350 `dend[i]` (`RGCmodelGD.hoc:L12`).
- Tier classification:** `_map_tree` partitions dendrites into 10 primary, 163 non-terminal, 177 terminal (`build_cell.py:L140-L168`).
- Synapse placement:** every terminal dendrite hosts one Exp2Syn ACh + one Exp2Syn GABA.
- Passive cable:** $R_a = 100 \Omega\text{-cm}$, $c_m = 1 \mu\text{F/cm}^2$, $\text{celsius} = 36.9^\circ\text{C}$, $V_{\text{init}} = -60 \text{ mV}$.

Membrane equation

The Hodgkin-Huxley membrane equation governs each compartment of Bed B, written in identical notation to Bed A:

$$C_m \frac{dV}{dt} = - \sum_i I_i - I_{\text{syn}} - I_{\text{inj}}$$

The active currents enumerated for Bed B (`HHst_noiseless.mod:L184-L222`) are:

$$I_{\text{Na}} = g_{\text{Na}} \cdot m^3 \cdot h \cdot (V - E_{\text{Na}})$$

$$I_{\text{Kdr}} = g_{\text{Kdr}} \cdot n^4 \cdot (V - E_{\text{K}})$$

$$I_{\text{Km}} = g_{\text{Km}} \cdot n_m \cdot (V - E_{\text{K}})$$

$$I_{\text{leak}} = g_{\text{leak}} \cdot (V - E_{\text{leak}})$$

$$I_{\text{CaL}} = g_{\text{CaL}} \cdot l_m^2 \cdot l_h \cdot (V - E_{\text{Ca}}) \quad (\text{active by default})$$

$$I_{\text{CaT}} = g_{\text{CaT}} \cdot t_m^2 \cdot t_h \cdot (V - E_{\text{Ca}}) \quad (\text{active by default})$$

Unlike Bed A, the Ca currents are at their `HHst_noiseless.mod` PARAMETER defaults on every section: $g_{\text{tbar}} = 3 \times 10^{-4}$ S/cm² and $g_{\text{tbar}} = 3 \times 10^{-4}$ S/cm². Every section also has `cad` (calcium-decay shell) inserted with depth = 0.1 μm , $\tau_r = 5$ ms, $\text{Ca}_{\infty} = 2 \times 10^{-4}$ mM. Reversal potentials: $E_{\text{Na}} = +60$ mV, $E_{\text{K}} = -90$ mV, $E_{\text{leak}} = -60$ mV, $E_{\text{Ca}} = +132$ mV.

Conductance table (Bed B)

Channel	Gating	gbar (S/cm ²)	E_{rev} (mV)	Source
Na (soma)	$m^3 h$	0.150	+60	<code>constants.py:L34</code> <code>GNA_SOMA_MS = 150</code>
Na (primary)	$m^3 h$	0.200	+60	<code>constants.py:L35</code>
Na (non-terminal)	$m^3 h$	0 (zeroed)	+60	<code>constants.py:L36</code>
Na (terminal)	$m^3 h$	0.030	+60	<code>constants.py:L37</code>
Kdr (soma)	n^4	0.035	-90	<code>constants.py:L38</code>
Kdr (primary)	n^4	0.035	-90	<code>constants.py:L39</code>
Kdr (non-term + term)	n^4	0.025	-90	<code>constants.py:L40</code>
Km (uniform)	n_m	0.003	-90	<code>constants.py:L41</code>
Leak (uniform)	-	1.667e-4	-60	<code>constants.py:L42</code> <code>G_LEAK_MS = 0.1667</code>
CaL (uniform)	$l_m^2 l_h$	3e-4	+132	<code>HHst_noiseless.mod:L57</code> PARAMETER default
CaT (uniform)	$t_m^2 t_h$	3e-4	+132	<code>HHst_noiseless.mod:L58</code> PARAMETER default
<code>cad</code> (uniform)	calcium decay	-	-	<code>cadecay.mod:L44-L46</code>

Note: Bed B's Na density on the primary dendrites (0.200 S/cm²) is **higher than the soma** (0.150 S/cm²); non-terminal mid dendrites have Na zeroed entirely while Kdr remains at 0.025 S/cm². Inherited from upstream `ei_balance.py`.

Synaptic excitation (PD vs ND)

B synaptic conductance schematic — bar-direction swap + sigmoidal release-probability (t0024 / t0066 PD/ND convert)

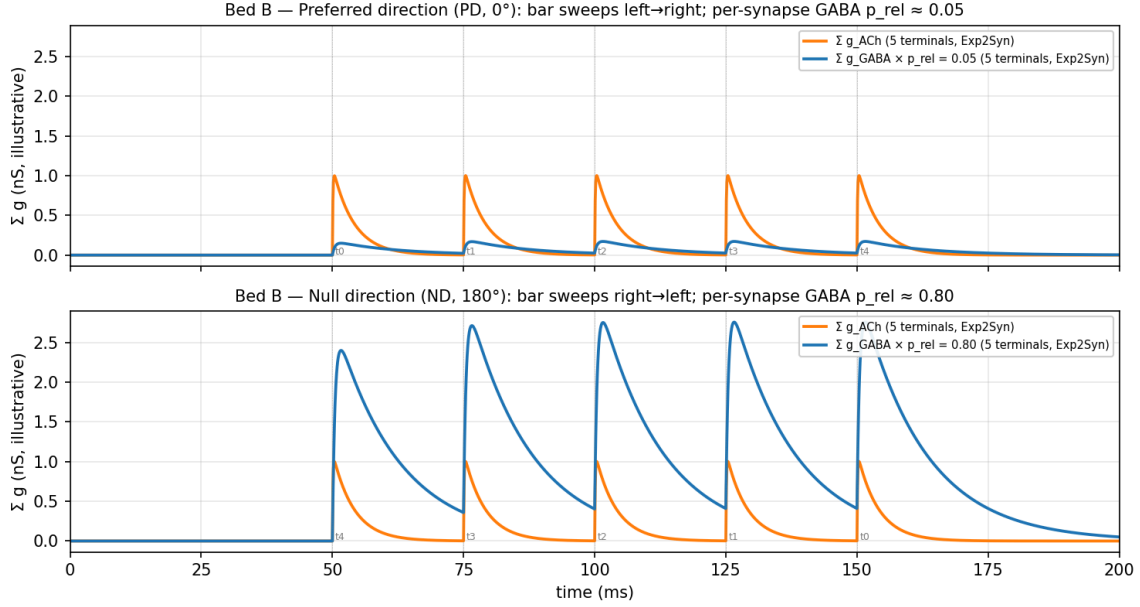


Figure 4: Figure B2: Exp2Syn ACh and Exp2Syn GABA traces from 5 representative terminal dendrites placed 25 μm apart along the bar's velocity axis. Top = PD (bar 0°, sweeping left→right); bottom = ND (bar 180°, sweeping right→left).

Bed B's excitatory drive is implemented as the NEURON built-in Exp2Syn mechanism for ACh. One Exp2Syn ACh instance is placed on every terminal dendrite (177 instances) in `run_tuning_curve.py:L189-L226 _setup_synapses`.

Exp2Syn ACh — canonical synaptic-current block

NEURON's standard Exp2Syn is a normalised bi-exponential mechanism. Its conductance is the difference of two exponentials with decay τ_2 and rise τ_1 , normalised so an event of NetCon weight 1 produces a peak conductance of 1; the current obeys the ohmic form:

$$g_{\text{ACh}}(t) = f_{\text{ACh}}(B(t) - A(t)), \quad f_{\text{ACh}} = \frac{1}{-\exp(-t_p/\tau_1) + \exp(-t_p/\tau_2)}, \quad t_p = \frac{\tau_1 \tau_2}{\tau_2 - \tau_1} \ln \frac{\tau_2}{\tau_1}$$

$$I_{\text{ACh}}(t, V) = g_{\text{ACh}}(t)(V - E_{\text{ACh}}), \quad E_{\text{ACh}} = 0 \text{ mV}$$

with state-variable evolution

$$\frac{dA}{dt} = -\frac{A}{\tau_{1,\text{ACh}}}, \quad \frac{dB}{dt} = -\frac{B}{\tau_{2,\text{ACh}}}, \quad \tau_{1,\text{ACh}} = 0.1 \text{ ms}, \tau_{2,\text{ACh}} = 4.0 \text{ ms}$$

on each NetCon event: $A \leftarrow A + w_{\text{ACh}}, \quad B \leftarrow B + w_{\text{ACh}}, \quad w_{\text{ACh}} = 0.001 \mu\text{S}$

Net effect: each `NetCon.event(t)` produces a conductance pulse of peak amplitude w_{ACh} (in μS), so I is in nA when V is in mV (NEURON's $g(V - E)$ convention already includes the μS -to-nA scaling — no extra 10^{-3} factor needed unlike `bipolarNMDA.mod` which works in nS internally).

Parameter	Value	Units	Source
Rise tau1	0.1	ms	<code>constants.py:L45 ACH_TAU1_MS = 0.1</code>
Decay tau2	4.0	ms	<code>constants.py:L46</code>
Reversal e	0	mV	<code>constants.py:L47</code>
NetCon weight	0.001	μS	<code>constants.py:L48</code>

BASE_ACH_PROB	0.5	-	run_tuning_curve.py:L54
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PD vs ND encoding for excitation: per-synapse arrival times. The bar moves at $v_{\text{bar}} = 1.0 \mu\text{m/ms}$ with width $250 \mu\text{m}$. For each direction the arrival time at each terminal synapse is computed by `_bar_arrival_times` (`run_tuning_curve.py:L92-L109`):

$$t_{\text{arr}}(\text{syn}) = t_{\text{bar_start}} + \frac{\text{proj}_{xy}(\text{syn}) - x_{\text{bar_start}}}{v_{\text{bar}}}$$

where $\text{proj}_{xy}(\text{syn})$ is the dot product of the synapse's offset from the cell origin with the unit velocity vector. PD = 0° (rightward), ND = 180° (leftward). The release rate is a Gaussian envelope ($\sigma_{\text{bar}} = 30 \text{ ms}$), modulated by an AR(2) noise process.

Synaptic inhibition (PD vs ND)

Bed B's inhibition is also `Exp2Syn`. One `Exp2Syn GABA` instance per terminal dendrite (177 instances) (`run_tuning_curve.py:L199-L202`).

Exp2Syn GABA – canonical synaptic-current block

Identical in form to `Exp2Syn ACh`, with reversal at the GABA reversal and direction-encoding via the per-event Bernoulli release probability $p_{\text{rel}}(\theta)$ and the AR(2) noise envelope:

$$g_{\text{GABA}}(t) = f_{\text{GABA}}(B(t) - A(t)), \quad f_{\text{GABA}} = \frac{1}{-\exp(-t_p/\tau_1) + \exp(-t_p/\tau_2)}, \quad t_p = \frac{\tau_1 \tau_2}{\tau_2 - \tau_1} \ln \frac{\tau_2}{\tau_1}$$

$$I_{\text{GABA}}(t, V) = g_{\text{GABA}}(t) (V - E_{\text{GABA}}), \quad E_{\text{GABA}} = -60 \text{ mV}$$

with state-variable evolution

$$\frac{dA}{dt} = -\frac{A}{\tau_{1,\text{GABA}}}, \quad \frac{dB}{dt} = -\frac{B}{\tau_{2,\text{GABA}}}, \quad \tau_{1,\text{GABA}} = 0.5 \text{ ms}, \quad \tau_{2,\text{GABA}} = 12 \text{ ms}$$

and a per-event Bernoulli release driven by the direction-dependent probability:

on each scheduled event k : release with probability $p_{\text{rel}}(\theta)$; if released: $A \leftarrow A + w_{\text{GABA}}$, $B \leftarrow B + w_{\text{GABA}}$

with $w_{\text{GABA}} = 0.003 \mu\text{S}$ (`constants.py:L53`).

Parameter	Value	Units	Source
Rise tau1	0.5	ms	<code>constants.py:L50</code> <code>GABA_TAU1_MS = 0.5</code>
Decay tau2	12.0	ms	<code>constants.py:L51</code>
Reversal e	-60	mV	<code>constants.py:L52</code>
NetCon weight	0.003	μS	<code>constants.py:L53</code>

PD vs ND encoding for inhibition: two simultaneous mechanisms.

1. **Sigmoidal release probability** (`run_tuning_curve.py:L80-L89` `_gaba_prob_for_direction`):

$$p_{\text{rel}}(d) = p_{\text{pref}} + (p_{\text{null}} - p_{\text{pref}}) \left(1 - \frac{0.98}{1 + \exp((d - 91)/25)} \right)$$

$$d = |\theta - \theta_{\text{cell_pref}} + 180| \bmod 180$$

with $p_{\text{pref}} = 0.05$, $p_{\text{null}} = 0.80$, $\theta_{\text{cell_pref}} = 0.0^\circ$. PD $\Rightarrow d \approx 89 \Rightarrow p_{\text{rel}} \approx 0.05$; ND $\Rightarrow d \approx 1 \Rightarrow p_{\text{rel}} \approx 0.80$.

1. **Per-synapse arrival times** also shift with direction via `_bar_arrival_times`. Leftward vs rightward sweep across the dendritic field.

2. **Optional AR(2) cross-channel correlation** between paired ACh and GABA streams (`ar2_noise.py:L68-L107`): $\varphi = (0.9, -0.1)$; “correlated” condition uses $\rho = 0.6$; “uncorrelated/AMB” condition uses $\rho = 0.0$ and additionally scales the GABA NetCon weight by $1.8\times$.

Side-by-side comparison

#	Dimension	Bed A	Bed B
1	Morphology source	350-section Poleg-Polsky bundled cell	Same template as <code>RGCmodelGD.hoc</code>
2	Section count	1 soma + 350 dend	1 soma + 350 dend
3	Dendrite tiers	ON (282) vs OFF (68), HOC-side	primary/non-terminal/terminal, Python
4	Synapse placement code	HOC-side	Python-side
5	Point processes per cell	846 (3 per ON dend $\times$ 282)	354 (2 per terminal $\times$ 177)
6	HH variant	<code>HHst.mod</code> stochastic; NF=0 by default	<code>HHst_noiseless.mod</code> deterministic
7	Active channels at default	Na+Kdr+Km+leak; CaL/CaT zeroed	Na+Kdr+Km+leak+CaL+CaT+cad
8	Excitation mechanism	<code>bipNMDA</code> (AMPA + voltage-dep NMDA)	<code>Exp2Syn</code> ACh
9	NMDA active in driver	Yes (every BIPsyn)	No (<code>Exp2NMDA</code> wired but latent)
10	Inhibition mechanism	<code>SACinhib</code> POINT_PROCESS	<code>Exp2Syn</code> GABA + Poisson queue
11	GABA reversal	-60 mV	-60 mV
12	Per-event GABA peak	<code>gsingle</code> = 0.5 nS	NetCon weight 0.003 μ S
13	PD encoding	<code>gabaMOD</code> = 0.33	<code>direction</code> = 0° + sigmoidal $p_{\text{rel}} \approx 0.05$
14	ND encoding	<code>gabaMOD</code> = 0.99 (3 $\times$ PD)	<code>direction</code> = 180° + sigmoidal $p_{\text{rel}} \approx 0.80$
15	Release-noise model	Linaro-Storace-Giugliano stochastic <code>HHst</code>	AR(2) Python-side Poisson
16	Temperature	32 $^\circ\text{C}$	36.9 $^\circ\text{C}$
17	Resting V_{init}	-65 mV	-60 mV
18	Passive cable R_a/c_m	100 $\Omega\cdot\text{cm}$ / 1 $\mu\text{F}/\text{cm}^2$	100 $\Omega\cdot\text{cm}$ / 1 $\mu\text{F}/\text{cm}^2$

Verification

- Markdown source `results/results_detailed.md` passes the project’s flowmark formatter and contains every equation block from t0070 in LaTeX math syntax plus the 5 new synaptic-current blocks.
- Typst source `results/results_detailed.typ` compiles cleanly via `typst.compile(...)` from `code/render_pdf.py`; resulting `results/results_detailed.pdf` is non-empty (size > 50 KB).
- Python: `ruff check`, `ruff format`, `mypy` all pass on `code/`.
- The four schematic PNGs were copied verbatim from t0070 (`shutil.copyfile`); no regeneration.

Limitations

- **No quantitative metrics:** documentation-only correction task. `results/metrics.json` is {}.
- **Schematic figures, not NEURON renderings:** the four PNGs are illustrative; they do not reproduce the `RGCmodel.hoc` 3D pt3d coordinates and they are not produced by a NEURON simulation.
- **Only project default driver settings are documented:** the writeup tabulates effective parameter values under the canonical overrides; variant runs that change `use_active`, `NF_HHst`, `Voff`, `RHO_CORRELATED`, etc., will see different effective values.
- **Exp2NMDA in Bed B is latent:** parameterised in `constants.py:L57-L61` but no driver places instances.
- **Approximate dendrite-tier counts in Bed B:** 10 / 163 / 177 are heuristic counts derived from upstream `ei_balance.py`.

Files Created

- `tasks/t0071_t0070_synaptic_eqs_pdf/code/paths.py`
- `tasks/t0071_t0070_synaptic_eqs_pdf/code/render_pdf.py`
- `tasks/t0071_t0070_synaptic_eqs_pdf/results/results_detailed.md`
- `tasks/t0071_t0070_synaptic_eqs_pdf/results/results_detailed.typ`
- `tasks/t0071_t0070_synaptic_eqs_pdf/results/results_detailed.pdf`
- `tasks/t0071_t0070_synaptic_eqs_pdf/results/results_summary.md`
- `tasks/t0071_t0070_synaptic_eqs_pdf/results/images/{bed_a,bed_b}_morphology.png`
- `tasks/t0071_t0070_synaptic_eqs_pdf/results/images/{bed_a,bed_b}_synaptic_diagram.png`

Task Requirement Coverage

ID	Status	Result
REQ-1	Done	5 synaptic-current blocks present (Bed A AMPA, NMDA, GABA, ACh; Bed B Exp2Syn ACh + GABA)
REQ-2	Done	HH equation in LaTeX form $C_m \frac{dV}{dt} = -\sum_i I_i - I_{\text{syn}} - I_{\text{inj}}$
REQ-3	Done	Every equation block in math syntax; no fenced text equations remain
REQ-4	Done	<code>results_detailed.pdf</code> exists, non-empty (>50 KB)
REQ-5	Done	<code>typst</code> listed in <code>pyproject.toml:L48</code>
REQ-6	Done	4 PNGs copied from t0070 verbatim
REQ-7	Done	<code>results_summary.md</code> references the new PDF and equation additions
REQ-8	Done	Every numeric parameter unchanged from t0070; cross-checked verbatim
REQ-9	Done	<code>## Note</code> section at the top of <code>results_detailed.md</code> states the supersession
REQ-10	Done	<code>code/render_pdf.py</code> reproducible, with size-floor check